

CHL5233: Quarto in-class assignment

Geographic coverage of the supplied country list

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This report describes the countries in the supplied course files, matches them using three-letter ISO codes, and summarizes their geographic distribution. The analysis counts countries; it does not estimate conflict exposure or health outcomes.

Read the source data

```
countries <- read.table(  
  "../data/raw/countries.txt", header = TRUE,  
  stringsAsFactors = FALSE, fileEncoding = "latin1"  
)  
dim(countries)
```

```
[1] 186    2
```

The country list contains **186 countries** and **2 columns**, with 186 distinct ISO codes. The supplied list contains 186 countries, although the paper title quoted in the assignment refers to 181 countries. This report uses the supplied list without dropping countries to match that title.

```
regions <- read.table(  
  "../data/raw/regions.txt", header = TRUE,  
  stringsAsFactors = FALSE, fileEncoding = "latin1"  
)  
dim(regions)
```

```
[1] 249    6
```

The region lookup contains **249 country or territory records** and **6 columns**. The raw file includes imperfectly encoded country names; ISO codes and geographic category labels are used for matching and summaries. The original files are preserved unchanged.

Create the analytical data set

```
analysis_data <- regions |>  
  semi_join(countries, by = c("country_code" = "ISO"))  
unmatched <- countries |>  
  anti_join(regions, by = c("ISO" = "country_code"))
```

All **186 countries** match successfully; there are **0 unmatched codes**. The analytical data cover **5 regions** and **17 sub-regions**.

Tabulate countries by region and sub-region

Each country contributes once to each table. Both sets of counts sum to **186**.

Countries by region

Table 1: Number of supplied study countries in each region.

Region	Count
Africa	54
Asia	46
Europe	40
Americas	35
Oceania	11

Countries by sub-region

Table 2: Number of supplied study countries in each sub-region.

Sub-region	Count
Sub-Saharan Africa	48
Latin America and the Caribbean	33
Western Asia	16
Southern Europe	13
South-eastern Asia	11
Eastern Europe	10
Northern Europe	10
Southern Asia	9
Western Europe	7
Northern Africa	6
Central Asia	5
Eastern Asia	5
Melanesia	4
Micronesia	3
Australia and New Zealand	2
Northern America	2
Polynesia	2

Africa contains the most included countries (**54**), while Oceania contains the fewest (**11**). Sub-Saharan Africa has the largest sub-region count (**48**).

Visualize the number of countries by region

The chart below addresses the assignment's overall goal of comparing the number of included countries across regions. Bars are ordered from the largest count at the top to the smallest at the bottom.

```
ggplot(analysis_data,
      aes(y = forcats::fct_infreq(region), fill = region)) +
  geom_bar(width = 0.65) +
  scale_y_discrete(limits = rev(region_counts$region)) +
  scale_fill_manual(values = region_colors, guide = "none") +
  scale_x_continuous(breaks = seq(0, 60, 10)) +
  labs(x = "Number of countries", y = "Region") +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank())
```

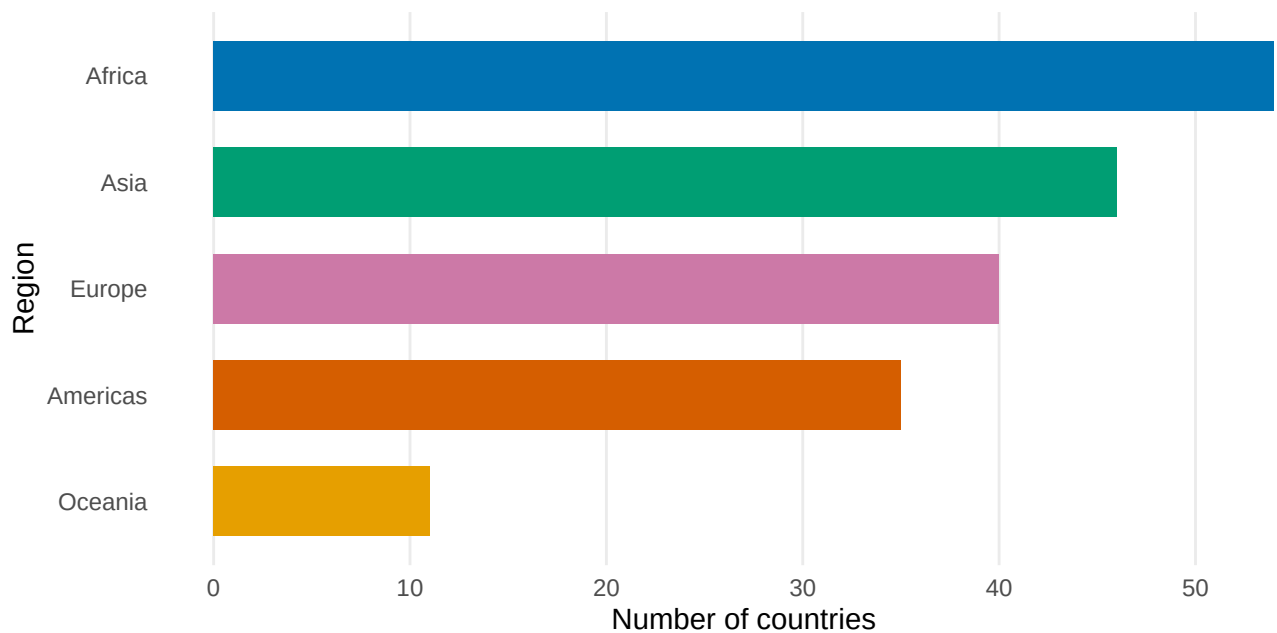


Figure 1: Geographic coverage of the 186 countries in the supplied list.

Interpretation. The country counts vary across regions. These bars describe the composition of the supplied study list, not regional population size, conflict prevalence, or the number of people affected by conflict.

Visualize the number of countries by sub-region

Each bar represents one sub-region. Colors identify its parent region. Counts run from largest at the top to smallest at the bottom; ties are ordered alphabetically. `geom_bar()` counts the country rows directly.

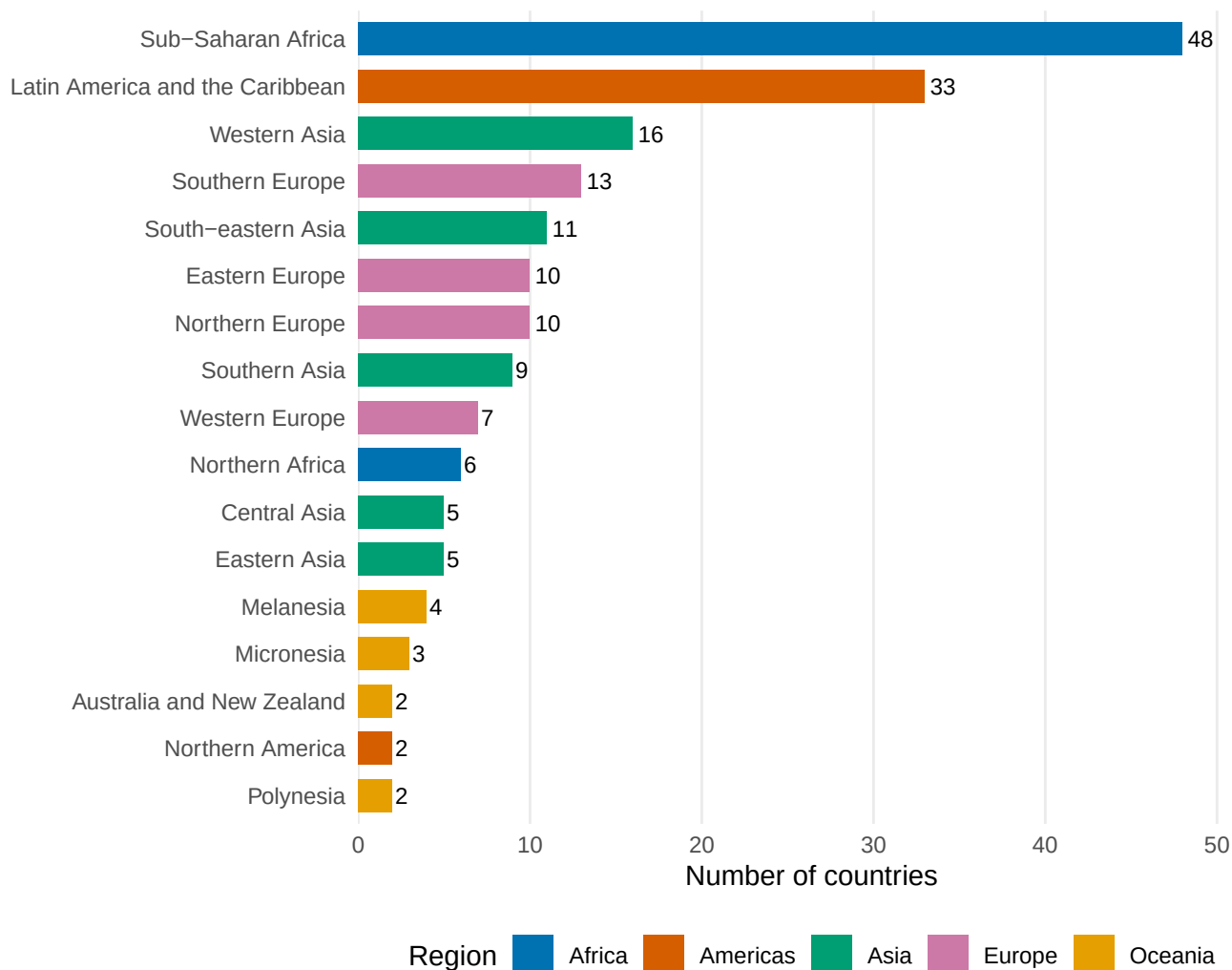


Figure 2: Countries by sub-region, ordered by count and colored by region.

Interpretation. Sub-Saharan Africa and Latin America and the Caribbean contain 48 and 33 included countries, respectively. The chart uses the regional classification provided in `regions.txt`; it does not redefine the sub-regions.

Data source. Course-provided `countries.txt` and `regions.txt`, associated with the assignment on *Implications of armed conflict for maternal and child health: A regression analysis of data from 181 countries for 2000-2019*.